

HW 2

Jacob Callets

① $x^3 - 7x^2 + 14x - 5 = 0$ for $e < 0.01$

A.) Bisection $x_0 = 0$ $x_1 = 1$

$x_2 = \frac{x_0 + x_1}{2}$ If $f(x_2) < 0$, swap x_0
else, swap x_1

i	x_0	x_1	x_2	$f(x_2)$	$e = x^{\text{new}} - x^{\text{old}} $
2	0	1	0.5	0.375	0.5
3	0	0.5	0.25	-1.921	0.25
4	0.25	0.5	0.375	-0.682	0.125
5	0.375	0.5	0.438	-0.127	0.063
6	0.438	0.5	0.469	0.129	0.031
7	0.438	0.469	0.454	0.007	0.015
8	0.438	0.454	0.446	-0.06	0.008

Note: Values are rounded to 3 decimals and used in that form for each calculation. I say $i=1$ is guess x_0 and x_1

B.) Newton - Raphson $x_0 = 0$

$$f(x) = x^3 - 7x^2 + 14x - 5$$

$$f'(x) = 3x^2 - 14x + 14$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

i	x_n	x_{n+1}	$f(x_{n+1})$	$e = x^{\text{new}} - x^{\text{old}} $
2	0	0.357	-0.849	0.357
3	0.357	0.447	-0.051	0.09
4	0.447	0.453	-0.0015	0.006

I say $i=1$ is guess x_0

Much faster than Bisection! Half the iterations